

JEE MAIN REVISION MASTER: PERIODIC TABLE AND PERIODICITY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 16 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Ionization Enthalpy & Electron Gain Enthalpy	15 PYQs	Periodic Trends
Standard	Atomic & Ionic Radii Trends	1 PYQs	Radii

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ionization Enthalpy & Electron Gain Enthalpy	$Z_{\text{eff}} = Z - \sigma, \quad \Delta_{\text{eg}}H(\text{Cl}) < \Delta_{\text{eg}}H(\text{F}) \quad (\text{Cl is most negative in periodic table})$
Atomic & Ionic Radii Trends	$\text{For isoelectronic: } r \propto \frac{1}{Z} \implies \text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ionization Enthalpy & Electron Gain Enthalpy (Very High Yield — 15 PYQs)

Core Principle: Periodic variations of ionization energy, exceptions due to half-filled/fully-filled subshells ($\text{N} > \text{O}$, $\text{Be} > \text{B}$), and electron gain enthalpy.

■ **Caution / JEE Pitfall:** Ionization enthalpy of Nitrogen is HIGHER than Oxygen due to stable half-filled $2p^3$ subshell ($\text{N}: 2s^2 2p^3 > \text{O}: 2s^2 2p^4$).

■ **Speed Trick:** Chlorine has MORE negative electron gain enthalpy than Fluorine because of compact size and high inter-electronic repulsion in $2p$ of Fluorine.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Match List-I with List-II: List-I (Element-Atomic number) List-II (Position in periodic table) (a) Ra – 88 (i) 4th period, 13th gr..." [Ans: N/A]
- **[2025 Shift] Q#2 (MCQ):** "Given below are the atomic masses of the elements: $\begin{matrix} \text{Element:} \\ \text{Li} \end{matrix}$ & Li & Na & Cl & K & Ca & Br & Sr & I & Ba ... " [Ans: N/A]
- **[2023 Shift] Q#1 (MCQ):** "In which one of the following pairs, both the elements does not have $(n-1)d^1 10ns^2$ configuration in its elementary state?" [Ans: N/A]

3.2 Atomic & Ionic Radii Trends (Standard Yield — 1 PYQs)

Core Principle: Covalent, van der Waals, and ionic radii trends across periods and groups, and size shrinkage in isoelectronic ions.

■ **Caution / JEE Pitfall:** Noble gas radii are measured as van der Waals radii, making them appear abnormally larger than adjacent halogens.

■ **Speed Trick:** Due to lanthanoid contraction (poor shielding of 4f electrons), 4d and 5d transition elements have nearly identical radii (e.g. Zr $\approx$ Hf).

Representative Source PYQs:

- [2018 Shift] Q#4 (MCQ): "Which of the following is the correct order of radius?..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II: List-I (Element-Atomic number) List-II (Position in periodic table) (a) Ra – 88 (i) 4th period, 13th group (b) Ga – 31 (ii) 6th period, 6th group (c) W – 74 (iii) 5th period, 10th group (d) Pd – 46 (iv) 7th period, 2nd group Choose the correct answer from the options given below.

- (A) a-iv, b-i, c-ii, d-iii
- (B) a-i, b-ii, c-iii, d-iv
- (C) a-iv, b-ii, c-iii, d-i
- (D) a-iii, b-iv, c-i, d-ii

Correct Answer: None

Solution Analysis: For finding the position of an element in the periodic table, we use its electronic configuration. (a) Ra, atomic number 88 Ra belongs to alkaline earth metals. Its outer configuration is: Ra:[Rn]7s² Highest shell number is 7, so it is in the 7th period. It has 2 valence electrons in the s-subshell, so it is in the 2nd group. So, Ra matches with (iv). (b) Ga, atomic number 31 Electronic configuration: Ga:[Ar]3d¹⁰4s²4p¹ Highest shell number is 4, so it is in the 4th period. It has configuration ns²np¹, so it belongs to group 13.

Benchmark Problem 2: KCET 2018 (Q#4)

Which of the following is the correct order of radius?

- (A) H⁻ > H > H⁺ [Correct]
- (B) Na⁺ > F⁻ > O²⁻
- (C) F⁻ > O²⁻ > Na⁺
- (D) Al³⁺ > Mg²⁺ > N³⁻

Correct Answer: A

Solution Analysis: Let's analyze each option step by step, keeping in mind that the effective radius of an ion or atom depends on its electron cloud and the nuclear charge. A useful principle is: • For species with similar electron configurations (isoelectronic species), a higher nuclear charge (more protons) pulls the electrons closer, resulting in a smaller radius. Now, let's evaluate each option: H⁻ > H > H⁺ H⁻ (hydride ion) has one extra electron compared to neutral hydrogen. The extra electron-electron repulsion tends to expand the electron cloud, making it larger than neutral H. Neutral H has a